Multiple Comparison Tests

Why post-hoc tests are needed, and Fisher's LSD (CD), Duncan's DMRT and Tukey's HSD with letter grouping.

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1 Why a post-hoc test?

A significant ANOVA (F) shows that **not all means are equal**, but not **which pairs** differ.

Number of pairs

$$N = \binom{v}{2} = \frac{v(v-1)}{2}$$

Inflated Type I error

$$1 - (1-\alpha)^N$$

FWER (family-wise error rate): probability of at least one Type I error across the whole family of comparisons.

2 Fisher's LSD = Critical Difference (CD)

Equal replications

$$\text{LSD} = t_{\alpha/2,\,edf}\sqrt{\frac{2\,MSE}{r}}$$

Unequal: $SE(d)$

$$\sqrt{MSE\left(\frac{1}{r_i} + \frac{1}{r_j}\right)}$$

Two steps: significant ANOVA first (**protected LSD**), then pairs. Means differ if $|\bar{T}_i - \bar{T}_j| > \text{LSD}$. Relatively **liberal**.

3 Duncan's Multiple Range Test (DMRT)

SE of a mean

$$SE(\bar{Y}) = \sqrt{\frac{MSE}{r}}$$

Shortest significant range

$$R_p = r_p \cdot SE(\bar{Y})$$

1. Rank the means
2. Get $SE(\bar{Y})$ and table r_p
3. Compare each difference with its R_p

p = number of ordered means spanned, including both ends (adjacent: $p = 2$). R_p grows with p. Popular in plant sciences.

4 Tukey's HSD

Honestly significant difference

$$\text{HSD} = q_{\alpha,(v,\,edf)}\sqrt{\frac{MSE}{r}}$$

Uses the **studentised range** q. Compares all pairs while **controlling the FWER**. One critical value for equal replication. Most **conservative** of the three; needs no preliminary ANOVA.

5 Compare the tests

	LSD / CD	DMRT	Tukey HSD
Critical value	One (t)	Varies with p (r_p)	One (q)
Error control	Good only for 2–3 treatments	Between the two	Controls FWER
Nature	Most liberal	More powerful than Tukey	Most conservative
Worked example	1.10	1.10 – 1.21	1.59

In the example, LSD and DMRT found every pair significant; Tukey did not declare differences of 1.5.

6 Other procedures

- **Dunnett:** every treatment vs one control
- **Bonferroni:** adjusts α for the number of tests
- **Scheffé:** all linear contrasts
- **SNK:** Student–Newman–Keuls

7 Presenting results: letter grouping

Treatment	Mean	Group
T3	24.6	**a**
T1	23.9	**ab**
T2	22.8	**b**

Same letter = not significantly different; **no common letter** = significantly different. A mean can carry two letters (overlapping groups). Line/underscore notation conveys the same information. (Illustrative values.)

! Common mistakes

- × Running many t tests without adjustment.
- × Unprotected LSD after a non-significant F.
- × Misreading "ab" as different from "a".

? Fill in the blanks

1. With 6 treatments there are _____ pairwise comparisons. [15]
2. At α = 0.05, the chance of at least one false positive in 3 independent comparisons is about _____. [0.14]
3. MSE = 8, r = 4 and t = 2.12 give an LSD of _____. [4.24]
4. Means marked “a” and “ab” are _____ different. [not significantly]
5. Comparing every new variety with one check variety uses _____ test. [Dunnett's]
6. Of LSD, DMRT and Tukey's HSD, the most conservative is _____. [Tukey's HSD]

Source: *Textbook of Agricultural Statistics*, Ch. 19 · Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul · KAU Me-LON

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